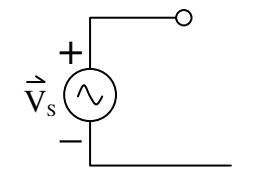


In step 3 of the 5-step PSA procedure, you are tasked with calculating numeric values for the parameters in the power system equivalent circuit drawn in step 2. The numeric values for the circuit parameters are calculated from the problem statement information and specifications given for the power system components. For each different type of power system component, the information given in standard specifications and examples of how to calculate equivalent circuit parameters are given below.

1. Ideal Source (also called an Infinite Bus)

Equivalent Circuit	Circuit Parameters
	$\vec{V}_s = V_s \angle \alpha_s$

Standard Specification Information:

- Rated Output Voltage (V_{rated} or V_{LLrated})
- Actual Operating Voltage (V or V_{LL})
- Voltage Source Phase Angle Information

Notes:

- In a single source system, assume the voltage source angle to be 0° . In a multi-source system, pick one voltage source to have a 0° angle, and then use whatever source phase angle information that is given to determine the other voltage source angles.
- For 3ϕ ideal sources, only L-N equivalent voltage can be used in the equivalent circuit, not L-L voltage.

Example #1a:

In a 1ϕ power system, three separate remotely located sources supply power to several distributed loads using a network of power cables. The three sources are all rated 380V. The first source operates at 383.25V, the second source operates at 384.14V, and the third source operates at 382.975V. The first source voltage leads the second source voltage by 2° , and the third source voltage lags the second source voltage by 0.5° . Calculate the equivalent circuit parameters for the three sources of this 1ϕ power system.

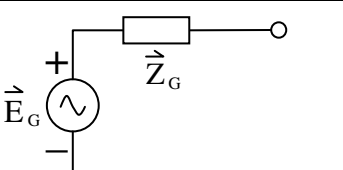
$$\vec{V}_{s1} = 383.25 \angle 0^\circ V, \quad \vec{V}_{s2} = 384.14 \angle -2^\circ V, \quad \vec{V}_{s3} = 382.975 \angle -2.5^\circ V$$

Example #1b:

In a 3ϕ power system, a Y-connected source is rated 960V and operates at 962.159V. Calculate the L-N equivalent circuit parameters for this source.

$$\vec{V}_s = \left(\frac{962.159}{\sqrt{3}} \right) \angle 0^\circ V = 555.5028 \angle 0^\circ V$$

2. Generator

Equivalent Circuit	Circuit Parameters
	$\vec{E}_G = E_G \angle \alpha_G$ $\vec{Z}_G = R_G + jX_G = Z_G \angle \theta_G$

Standard Specification Information:

- Rated Maximum Apparent Power Output (S_{rated} or $S_{3\phi\text{rated}}$)
- Rated Output Voltage (V_{rated} or V_{LLrated})
- No-Load Operating Voltage (E_G or E_{LLG})
- Stator Winding Impedance (Z_G and θ_G) or (R_G and X_G) or (X_G in % and X/R) or (Z_G in % and X/R)
- Voltage Source Phase Angle Information
- Three Phase Connection Scheme (Y or Δ)

Notes:

- In a single source system, assume the internal voltage source angle to be 0° . In a multi-source system, pick one voltage source to have a 0° angle, and then use whatever source angle information that is given to determine the other voltage source angles.
- The specified no-load operating voltage should be used to determine the magnitude of the internal voltage source of the equivalent circuit.
- For 3ϕ generators, only L-N equivalent voltage can be used in the equivalent circuit, not L-L voltage.
- The specified X/R can be used to determine either R_G or X_G assuming that one of the quantities is already known. The specified X/R can also be used to determine the generator winding impedance angle θ_G by simply taking the inverse tangent of the specified X/R.

$$R_G = X_G / (X / R) \quad \text{or} \quad X_G = R_G (X / R) \quad \text{or} \quad \theta_G = \tan^{-1} (X / R)$$

- If the stator winding impedance is given using percent per unit quantities (X_G in %) or (Z_G in %), then the following special conversion is required to obtain the same quantity in actual Ohms.

$$(base \Omega) = \frac{(rated \text{ KV})^2}{(rated \text{ MVA})} \quad \text{then} \quad X_G = \left(\frac{X_G \text{ in \%}}{100} \right) (base \Omega) \quad \text{or} \quad Z_G = \left(\frac{Z_G \text{ in \%}}{100} \right) (base \Omega)$$

- For 3ϕ generators, the calculated Z_G value in the equivalent circuit must be divided by three if the generator is Δ -connected.
- Generator output voltage and output powers are defined at the equivalent circuit external connection node, not at the equivalent circuit internal voltage source.

Example #2a:

A generator supplies a 1ϕ power system. The generator is rated 560KVA, 240V and has a stator winding reactance of 19.35% with a 175 X/R ratio. The generator no-load voltage is adjusted to operate at 255.75V. Calculate the equivalent circuit parameters for the generator.

$$\vec{E}_G = 255.75 \angle 0^\circ \text{ V}$$

$$X_G = \left(\frac{X_G \text{ in \%}}{100} \right) \left[\frac{(rated \text{ KV})^2}{(rated \text{ MVA})} \right] = \left(\frac{19.35\%}{100} \right) \left[\frac{(0.240)^2}{0.560} \right] = 0.019903 \Omega$$

$$R_G = X_G / (X / R) = (0.019903 \Omega) / (175) = 0.000114 \Omega$$

$$\vec{Z}_G = R_G + jX_G = 0.019903 \angle 89.673^\circ \Omega$$

Example #2b:

In a 3ϕ power system, a Δ -connected generator is rated 5MVA, 960V and has a stator winding impedance of 23.6% with a 105 X/R ratio. The generator is adjusted to operate with a no-load voltage of 967.952V. Calculate the L-N equivalent circuit parameters for this generator.

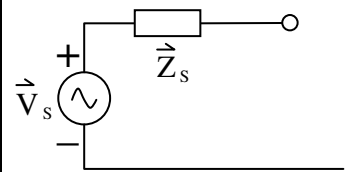
$$\bar{E}_G = \left(\frac{967.952}{\sqrt{3}} \right) \angle 0^\circ V = 558.84735 \angle 0^\circ V$$

$$Z_G = \frac{1}{3} \left(\frac{Z_G \text{ in } \%}{100} \right) \left[\frac{(\text{rated KV})^2}{(\text{rated MVA})} \right] = \frac{1}{3} \left(\frac{23.6\%}{100} \right) \left[\frac{(0.960)^2}{5} \right] = 0.0145 \Omega$$

$$\theta_G = \tan^{-1} (X / R) = \tan^{-1} (105) = 89.454^\circ$$

$$\bar{Z}_G = Z_G \angle \theta_G = 0.0145 \angle 89.454^\circ \Omega$$

3. Electric Utility Tie

Equivalent Circuit	Circuit Parameters
	$\bar{V}_s = V_s \angle \alpha_s$ $\bar{Z}_s = R_s + jX_s = Z_s \angle \theta_s$

Standard Specification Information:

- Rated Output Voltage (V_{rated} or V_{LLrated})
- No-Load Operating Voltage (V_s or V_{LLS})
- Source Impedance (R_s and X_s) or (Z_s and θ_s) or (Z_s and X/R) or (S_{SC} or $S_{3\phi\text{SC}}$ and X/R)
- Voltage Source Phase Angle Information

Notes:

- In a single source system, assume the internal voltage source angle to be 0° . In a multi-source system, pick one voltage source to have a 0° angle, and then use whatever source angle information that is given to determine the other voltage source angles.

- If the source impedance is given using short circuit apparent power and X/R ratio (S_{SC} and X/R), then the following conversion is required to obtain the same information in (Z_s and θ_s) format.

$$Z_s = \frac{V_{\text{rated}}^2}{S_{\text{SC}}} \quad \text{or} \quad Z_s = \frac{V_{\text{LLrated}}^2}{S_{3\phi\text{SC}}} \quad \text{and} \quad \theta_s = \tan^{-1} (X / R)$$

- For 3ϕ electric utility ties, only L-N equivalent voltage can be used in the equivalent circuit, not L-L voltage.
- Electric utility tie output voltage and output powers are defined at the equivalent circuit external connection node, not at the equivalent circuit internal voltage source.

Example #3a:

A 3ϕ power system is sourced by a tie with the electric utility company. The electric utility company tie is rated 13.8KV and operates with a no-load voltage of 13,812.15V. The tie has a source impedance of 3.0917Ω and 12.53 X/R ratio Calculate the equivalent circuit parameters for the electric utility company tie.

$$\bar{V}_s = \left(\frac{13,812.15}{\sqrt{3}} \right) \angle 0^\circ V = 7974.4485 \angle 0^\circ V$$

$$\theta_s = \tan^{-1} (X / R) = \tan^{-1} (12.53) = 85.437^\circ$$

$$\bar{Z}_s = Z_s \angle \theta_s = 3.0917 \angle 85.437^\circ \Omega$$

Example #3b:

A 1 ϕ power system is sourced by a tie with the electric utility company. The electric utility company tie is rated 8KV with a short circuit KVA of 1250 and 16.25 X/R ratio. The electric utility company tie operates with a no-load voltage of 8050V. Calculate the equivalent circuit parameters for the electric utility company tie.

$$\vec{V}_S = 8050 \angle 0^\circ \text{ V}$$

$$Z_S = \frac{V_{rated}^2}{S_{SC}} = \frac{(8 \times 10^3)^2}{1250 \times 10^3} = 51.2 \, \Omega, \quad \theta_S = \tan^{-1}(X/R) = \tan^{-1}(16.25) = 86.479^\circ$$

$$\vec{Z}_S = Z_S \angle \theta_S = 51.2 \angle 86.479^\circ \, \Omega$$

Example #3c:

In a 3 ϕ power system, an electric utility company tie is rated 115KV with a short circuit MVA of 1746.6 and a 9.68046 X/R ratio. Calculate the L-N equivalent circuit parameters for this electric utility company tie.

$$\vec{V}_S = \left(\frac{115,000}{\sqrt{3}} \right) \angle 0^\circ \text{ V} = 66,395.28096 \angle 0^\circ \text{ V}$$

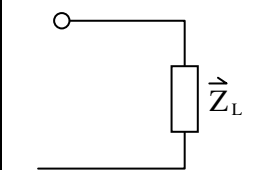
Rated voltage was used since the no-load operating voltage was not given

$$Z_S = \frac{V_{LLrated}^2}{S_{3\phi SC}} = \frac{(115 \times 10^3)^2}{1746.6 \times 10^6} = 7.571854 \, \Omega$$

$$\theta_S = \tan^{-1}(X/R) = \tan^{-1}(9.68046) = 84.102^\circ$$

$$\vec{Z}_S = Z_S \angle \theta_S = 7.571854 \angle 84.102^\circ \, \Omega$$

4. Load

Equivalent Circuit	Circuit Parameters
	$\vec{Z}_L = R_L + jX_L = Z_L \angle \theta_L$

Standard Specification Information:

- Impedance (Z_L and θ_L) or (R_L and X_L)
- Three Phase Connection Scheme (Y or Δ)

Notes:

- For 3 ϕ loads, the Z_L value in the equivalent circuit must be divided by three if the load is Δ -connected.

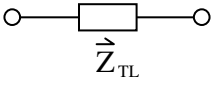
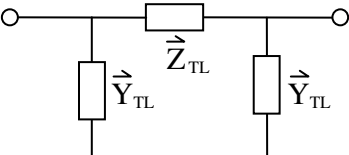
Example #4a:

In a 3 ϕ power system, a Δ -connected load with per phase impedance $60 \angle 25^\circ \Omega$ and a Y-connected load with per phase impedance $100 \angle -15^\circ \Omega$ are both connected to the same bus. Calculate the L-N equivalent circuit parameters for each of these loads and the net L-N equivalent impedance of both loads combined in the equivalent circuit.

$$\vec{Z}_{L1} = \frac{1}{3}(60 \angle 25^\circ \Omega) = 20 \angle 25^\circ \, \Omega, \quad \vec{Z}_{L2} = 100 \angle -15^\circ \, \Omega$$

$$\vec{Z}_{eq} = (\vec{Z}_{L1}^{-1} + \vec{Z}_{L2}^{-1})^{-1} = 17.236143 \angle 18.639^\circ \, \Omega$$

5. Transmission Line

Equivalent Circuit		Circuit Parameters
Series Impedance Model 	Pi Model 	$\bar{Z}_{TL} = \bar{z} \ell \quad (\Omega)$ $\bar{Y}_{TL} = \frac{1}{2} \bar{y} \ell \quad (\text{S})$

Standard Specification Information:

- Distributed Series Impedance (Ohms per unit length)
- Distributed Shunt Admittance (Mhos per unit length)
- Transmission Line Length
- Rated Maximum Conductor Current (also called conductor ampacity)

Notes:

- Transmission lines can also be specified using detailed physical construction data or other specially defined impedance parameters.
- In 1 ϕ transmission lines, there are two physical conductors (x and y) and both conductors must be accounted for in the series impedance circuit parameter calculation.

$$\bar{z} = \bar{z}_x + \bar{z}_y \quad \text{then} \quad \bar{Z}_{TL} = \bar{z} \ell$$
- In 3 ϕ transmission lines, there can be either three physical conductors (ABC) or four physical conductors (ABCN). However the equivalent circuit for a 3 ϕ transmission line is a L-N equivalent circuit, and only one of the phase conductors is modeled with the assumption that the same physical conductor type is used in all three of the phase conductors.
- The factor of one half in the equivalent circuit shunt admittance calculation is required to account for the total shunt admittance being split into two equal parts for the equivalent circuit model.

Example #5a:

In a 1 ϕ power system, a 1.15 mile transmission line uses a cable with a distributed series impedance of $1.2 \angle 85^\circ \Omega / 1000\text{ft}$ in one conductor and uses a cable with a distributed series impedance of $0.005 \angle 70^\circ \Omega / \text{ft}$ in the other conductor. The conductors have a distributed shunt admittance of $2.5 \angle 90^\circ \mu\text{S}$ per mile between them. Calculate the pi model equivalent circuit parameters for this transmission line.

$$\ell = 1.15 \text{ mi}$$

$$\bar{z}_x = (1.2 \angle 85^\circ \Omega / kft) \left(\frac{5.280 \text{ kft}}{1 \text{ mi}} \right) = 6.336 \angle 85^\circ \Omega / \text{mi}$$

$$\bar{z}_y = (0.005 \angle 70^\circ \Omega / \text{ft}) \left(\frac{5280 \text{ ft}}{1 \text{ mi}} \right) = 26.4 \angle 70^\circ \Omega / \text{mi}$$

$$\bar{z} = \bar{z}_x + \bar{z}_y = 32.561426 \angle 72.887^\circ \Omega / \text{mi}, \quad \bar{Z}_{TL} = \bar{z} \ell = 37.445640 \angle 72.887^\circ \Omega$$

$$\bar{Y}_{TL} = \frac{1}{2} \bar{y} \ell = \frac{1}{2} (2.5 \times 10^{-6} \angle 90^\circ \text{ S} / \text{mi}) (1.15 \text{ mi}) = 1.4375 \times 10^{-6} \angle 90^\circ \text{ S}$$

Example #5b:

In a 3 ϕ power system, a 1300 foot underground power cable uses four conductors, with each conductor having a distributed series impedance of $0.0039 \angle 65^\circ \Omega$ per meter. The cable has a distributed shunt admittance of

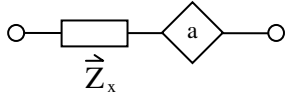
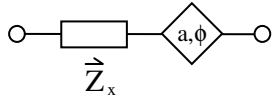
0.00125∠90°mΩ per kilometer for each phase conductor. Calculate the pi model L-N equivalent circuit parameters for this underground power cable.

$$\ell = (1300 \text{ ft}) \left(\frac{0.3048 \text{ m}}{1 \text{ ft}} \right) = 396.24 \text{ m}$$

$$\bar{Z}_{TL} = \bar{Z} \ell = (0.0039 \angle 65^\circ \Omega / \text{m}) (396.24 \text{ m}) = 1.545336 \angle 65^\circ \Omega$$

$$\bar{Y}_{TL} = \frac{1}{2} \bar{y} \ell = \frac{1}{2} (0.00125 \times 10^{-3} \angle 90^\circ \text{ S} / \text{km}) \left(\frac{1 \text{ km}}{1000 \text{ m}} \right) (396.24 \text{ m}) = 2.4765 \times 10^{-7} \angle 90^\circ \text{ S}$$

6. Transformer

Equivalent Circuit		Circuit Parameters
1φ XFMR Model 	3φ XFMR Model 	$a = V_{1rated} / V_{2rated}$ $\bar{Z}_x = R_1 + jX_1 = (r_1 + jx_1) + a^2 (r_2 + jx_2) = Z_x \angle \theta_x$ <i>divide $(r_x + jx_x)$ by 3 if the side is Δ</i> $\phi = 0^\circ, \pm 30^\circ, \text{ or } \pm 60^\circ$

Standard Specification Information:

- Rated Maximum Apparent Power Output (S_{rated} or $S_{3\phi rated}$)
- Rated Nominal Winding Voltages (V_{1rated} / V_{2rated} or $V_{LL1rated} / V_{LL2rated}$)
- Winding Impedances (r_1, x_1, r_2, x_2) or (R_1 and X_1) or (Z_x in % and X/R)
- Three Phase Connection Scheme (Y or Δ or Δ_{CBA}) of Each Side

Notes:

- The parameter a in the transformer model is called the TTR of the transformer, which stands for Transformer Turns Ratio, and it is calculated by taking the ratio of the specified rated primary side voltage to the specified rated secondary side voltage.
- If the winding impedances are given in (r_1, x_1, r_2, x_2) format with units of % instead of Ω, then the following special conversion is required to obtain the equivalent circuit impedance in units of Ω.

$$\bar{Z}_x = \left(\frac{r_1 + jx_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] + a^2 \left(\frac{r_2 + jx_2 \text{ in } \%}{100} \right) \left[\frac{(\text{rated secondary side KV})^2}{(\text{rated MVA})} \right]$$

divide $(r_1 + jx_1)$ by 3 if the primary side is Δ, divide $(r_2 + jx_2)$ by 3 if the secondary side is Δ

- If the winding impedances are given in (R_1, X_1) format with units of % instead of Ω, then the following special conversion is required to obtain the equivalent circuit impedance in units of Ω.

$$\bar{Z}_x = \left(\frac{R_1 + jX_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right]$$

- If the winding impedances are given in (Z_x in % and X/R) format, then the following special conversion is required to obtain the equivalent circuit impedance in units of Ω.

$$\bar{Z}_x = \left(\frac{Z_x \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] \angle \tan^{-1}(X / R)$$

- The parameter ϕ in the three phase transformer model is a special quantity that models a primary to secondary phase shift that occurs in voltages and currents of a three phase transformer. The value of this parameter ϕ is determined by the three phase transformer primary-secondary winding connection schemes used in the physical construction of the three phase transformer. The following chart should be used to obtain the appropriate value of the ϕ parameter in the three phase transformer L-N equivalent model.

$$\begin{array}{lll} \phi = 0^\circ \text{ if Y-Y or } \Delta-\Delta \text{ or } \Delta_{CBA}-\Delta_{CBA} & \phi = -30^\circ \text{ if Y-}\Delta \text{ or } \Delta_{CBA}\text{-Y} & \phi = -60^\circ \text{ if } \Delta_{CBA}\text{-}\Delta \\ & = +30^\circ \text{ if Y-}\Delta_{CBA} \text{ or } \Delta\text{-Y} & = +60^\circ \text{ if } \Delta\text{-}\Delta_{CBA} \end{array}$$

Example #6a:

A transformer in a 1 ϕ power system is rated 50KVA, 480V/8KV, has primary winding resistance and reactance of 0.01 Ω and 0.11 Ω , and has secondary winding resistance and reactance of 0.21 Ω and 2.85 Ω . Calculate the equivalent circuit parameters for this transformer.

$$a = \frac{480}{8000} = 0.06, \quad \bar{Z}_x = (r_1 + jx_1) + a^2(r_2 + jx_2) = (0.01 + j0.11) + a^2(0.21 + j2.85) = 0.12074 \angle 84.889^\circ \Omega$$

Example #6b:

A transformer in a 1 ϕ power system is rated 125KVA, 8KV/240V, 12% impedance, X/R ratio of 15. Calculate the equivalent circuit parameters for this transformer.

$$a = \frac{8000}{240} = 33.33$$

$$\bar{Z}_x = \left(\frac{Z_x \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] \angle \tan^{-1}(X/R) = \left(\frac{12}{100} \right) \left[\frac{(8)^2}{(0.125)} \right] \angle \tan^{-1}(15) = 61.44 \angle 86.186^\circ \Omega$$

Example #6c:

A transformer in a 3 ϕ power system is Y- Δ -connected and is rated 200MVA, 230KV/34.5KV, primary winding having 0.2% resistance and 3% reactance, and secondary winding having 1% resistance and 9% reactance. Calculate the L-N equivalent circuit parameters for this transformer.

$$a = \frac{230}{34.5} = 6.66, \quad \phi = -30^\circ$$

$$\bar{Z}_x = \left(\frac{r_1 + jx_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] + a^2 \left(\frac{r_2 + jx_2 \text{ in } \%}{100} \right) \left[\frac{(\text{rated secondary side KV})^2}{(\text{rated MVA})} \right]$$

$$\bar{Z}_x = \left(\frac{0.2 + j3}{100} \right) \left[\frac{(230)^2}{(200)} \right] + a^2 \left(\frac{1 + j9}{100 \cdot 3} \right) \left[\frac{(34.5)^2}{(200)} \right] = 15.932573 \angle 84.920^\circ \Omega$$

Example #6d:

A transformer in a 1 ϕ power system is rated 100KVA, 8KV/240KV, 2% resistance, X/R ratio of 17. Calculate the equivalent circuit parameters for this transformer.

$$a = \frac{8000}{240} = 33.33, \quad X_1 = R_1 (X/R) = (2\%)(17) = 34\%$$

$$\bar{Z}_x = \left(\frac{R_1 + jX_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] = \left(\frac{2 + j34}{100} \right) \left[\frac{(8)^2}{(0.100)} \right] = 2.179761 \angle 86.634^\circ \Omega$$